



GIS Project: Frozen Yogurt

Coming Soon: Your Perfect Scoop

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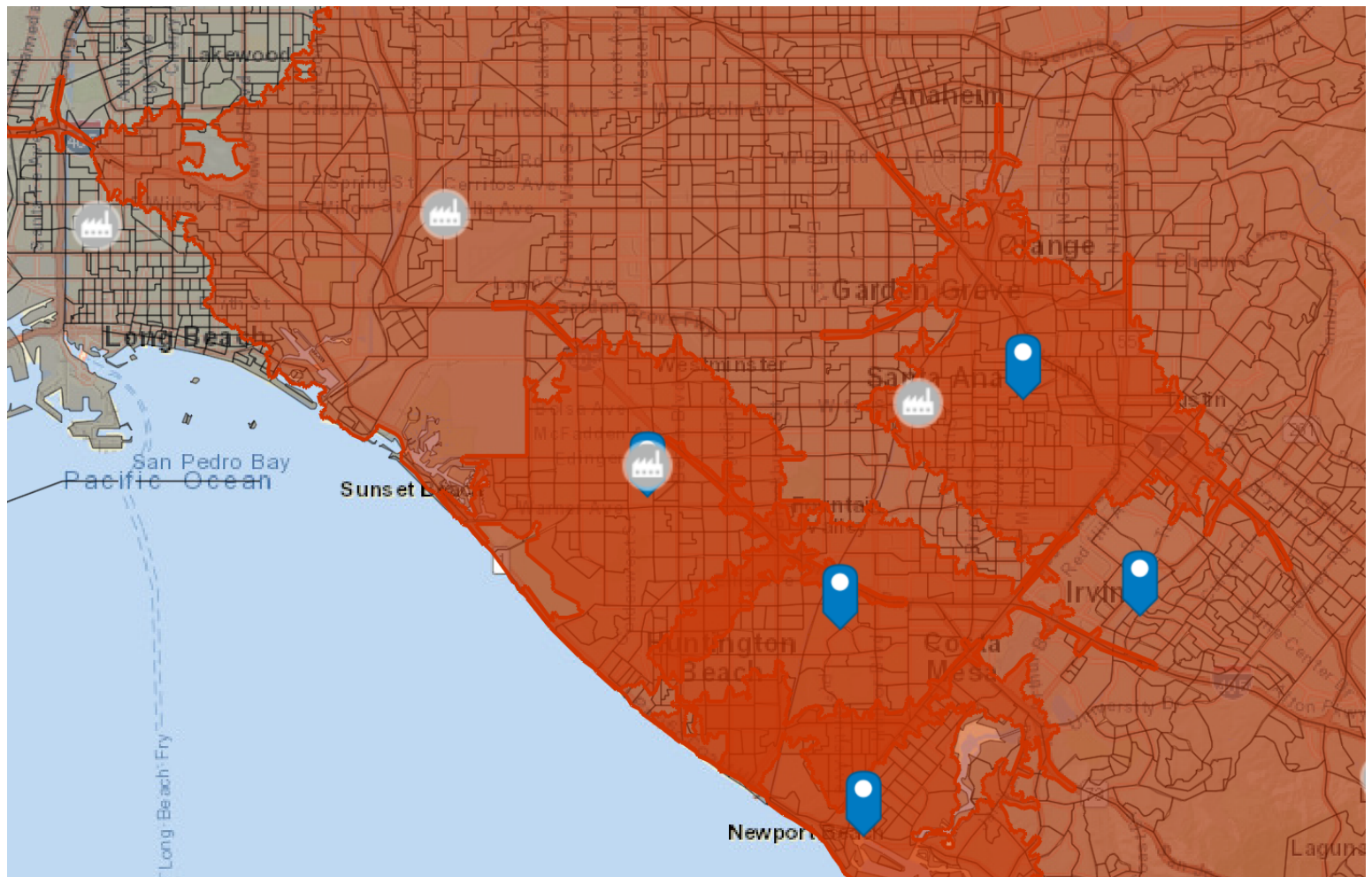
Introduction

The purpose of this project was to identify the best location for a frozen yogurt business we wanted to pursue. We did this by analyzing potential sites and competitors. Most frozen yogurt shops are concentrated in Irvine, but we found relatively few in nearby cities such as Santa Ana, Newport Beach, and Costa Mesa, which became the focus of our analysis.

We chose these locations because Irvine is a well-known area for dining and shopping, including the Irvine Spectrum Mall.

Its popularity and proximity to the beach provide an opportunity to attract both locals and tourists. Establishing a frozen yogurt shop in these areas could increase customer visits and sales, making them promising options for expanding our business.

This analysis allowed us to evaluate market demand, competition, and accessibility to determine which location offers the greatest potential for a successful frozen yogurt shop.



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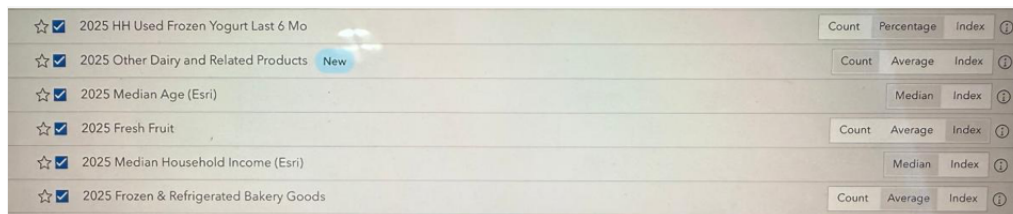
5 km Powered by Esri

Smart Map Search

The area we focused on included a 10-minute driving radius around three potential locations near Irvine, an area with a wide range of existing frozen yogurt competitors. This allowed us to explore multiple options and analyze nearby areas to determine the best site for a new shop. The three locations we considered for expansion were Santa Ana, Newport Beach, and Costa Mesa.

To evaluate potential demand, we selected variables commonly associated with frozen yogurt. Using Smart Map, we analyzed the percentage of people who had purchased frozen yogurt in the last six months, other dairy and related product purchases, median age, fresh fruit purchases often used as toppings, median income, and frozen or refrigerated bakery purchases. These factors helped us identify areas with the strongest potential for a successful frozen yogurt shop.

Variable List



☒ 2025 HH Used Frozen Yogurt Last 6 Mo	Count	Percentage	Index	ⓘ
☒ 2025 Other Dairy and Related Products New	Count	Average	Index	ⓘ
☒ 2025 Median Age (Esri)		Median	Index	ⓘ
☒ 2025 Fresh Fruit	Count	Average	Index	ⓘ
☒ 2025 Median Household Income (Esri)		Median	Index	ⓘ
☒ 2025 Frozen & Refrigerated Bakery Goods	Count	Average	Index	ⓘ

These variables matter for a frozen yogurt shop because they help identify whether people in the area are likely to buy frozen yogurt and related products. The percentage of people who have frozen yogurt in the last 6 months shows residents already enjoy frozen yogurt. The percentage of other dairy and related products indicates that people who often buy dairy items may also purchase frozen yogurt. Median age matters because younger adults, teenagers, and families tend

to visit frozen yogurt shops. The index of fresh fruit used as toppings shows that people who enjoy fresh fruit may prefer frozen yogurt as a healthier dessert option. Median household income affects spending habits, as people with higher incomes may be more willing to buy frozen yogurt. The percentage of frozen and refrigerated bakery goods shows interest in desserts and snacks, suggesting people may be interested in frozen yogurt.



Retail Market Potential

Costa Mesa
Capri Ln, Costa Mesa, California, 92626
Drive time: 10 minute radius



Demographic Summary	2025	2030
Population	186,879	187,788
Population 18+	153,271	155,602
Households	71,509	73,081
Median Household Income	\$115,093	\$129,557

Product/Consumer Behavior	Expected Number of Adults or HHs	Percent	MPI
Apparel (Adults)			
Bought Men's Clothing Last 12 Mo	96,495	63.0%	99
Bought Women's Clothing Last 12 Mo	77,738	50.7%	97
Bought Shoes Last 12 Mo	116,082	75.7%	100
Bought Fine Jewelry Last 12 Mo	34,695	22.6%	103
Bought Watch Last 12 Mo	20,039	13.1%	102
Automobiles (Households)			
HH Owns or Leases 1+ Vehicles	61,973	86.7%	97
HH Bought or Leased New Vehicle Last 12 Mo	6,840	9.6%	113
Automotive Aftermarket (Adults)			
Bought Gasoline Last 6 Mo	131,310	85.7%	96
Bought or Changed Motor Oil Last 12 Mo	74,095	48.3%	89
Had Vehicle Tune-Up Last 12 Mo	40,195	26.2%	114
Beverages (Adults)			
Drank Non-Diet (Regular) Cola Last 6 Mo	53,208	34.7%	91
Drank Beer or Ale Last 6 Mo	61,250	40.0%	107

Data Note: An MPI (Market Potential Index) measures the relative likelihood of the adults or households in the specified trade area to exhibit certain consumer behavior or purchasing patterns compared to the U.S. An MPI of 100 represents the U.S. average.

Source: These data are based upon national propensities to use various products and services, applied to local demographic composition. Usage data were collected by MRI-Simmons in a nationally representative survey of U.S. households. Esri forecasts for 2025 and 2030.

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Costa Mesa: Retail Market Potential

Retail Market Potential

Newport Beach

Drive time: 10 minute radius



Demographic Summary	2025	2030
Population	117,491	117,048
Population 18+	98,192	98,402
Households	50,005	50,623
Median Household Income	\$133,632	\$153,084

Product/Consumer Behavior	Expected Number of Adults or HHs	Percent	MPI
Apparel (Adults)			
Bought Men's Clothing Last 12 Mo	61,911	63.0%	99
Bought Women's Clothing Last 12 Mo	50,383	51.3%	98
Bought Shoes Last 12 Mo	75,180	76.6%	101
Bought Fine Jewelry Last 12 Mo	21,707	22.1%	101
Bought Watch Last 12 Mo	12,602	12.8%	100
Automobiles (Households)			
HH Owns or Leases 1+ Vehicles	43,334	86.7%	97
HH Bought or Leased New Vehicle Last 12 Mo	4,564	9.1%	108
Automotive Aftermarket (Adults)			
Bought Gasoline Last 6 Mo	84,353	85.9%	96
Bought or Changed Motor Oil Last 12 Mo	46,609	47.5%	88
Had Vehicle Tune-Up Last 12 Mo	25,123	25.6%	111
Beverages (Adults)			
Drank Non-Diet (Regular) Cola Last 6 Mo	31,317	31.9%	84
Drank Beer or Ale Last 6 Mo	40,013	40.8%	109

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Newport Beach: Retail Market Potential

Retail Market Potential

Santa Ana
E 4th St, Santa Ana, California, 92701
Drive time: 10 minute radius



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Demographic Summary	2025	2030
Population	301,067	302,766
Population 18+	234,041	240,700
Households	82,883	85,276
Median Household Income	\$89,997	\$102,923

Product/Consumer Behavior	Expected Number of Adults or HHs	Percent	MPI
Apparel (Adults)			
Bought Men's Clothing Last 12 Mo	150,551	64.3%	101
Bought Women's Clothing Last 12 Mo	120,017	51.3%	98
Bought Shoes Last 12 Mo	176,137	75.3%	99
Bought Fine Jewelry Last 12 Mo	59,567	25.4%	116
Bought Watch Last 12 Mo	33,463	14.3%	111
Automobiles (Households)			
HH Owns or Leases 1+ Vehicles	69,931	84.4%	94
HH Bought or Leased New Vehicle Last 12 Mo	7,318	8.8%	104
Automotive Aftermarket (Adults)			
Bought Gasoline Last 6 Mo	194,461	83.1%	93
Bought or Changed Motor Oil Last 12 Mo	118,613	50.7%	94
Had Vehicle Tune-Up Last 12 Mo	54,988	23.5%	102
Beverages (Adults)			
Drank Non-Diet (Regular) Cola Last 6 Mo	102,241	43.7%	115
Drank Beer or Ale Last 6 Mo	89,436	38.2%	103

Data Note: An MPI (Market Potential Index) measures the relative likelihood of the adults or households in the specified trade area to exhibit certain consumer behavior or purchasing patterns compared to the U.S. An MPI of 100 represents the U.S. average.

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Santa Ana: Retail Market Potential

FROYO

Add data

Create maps

Define areas

Run analysis

Share results

Search for places or areas

50 mi

Competitive Analysis

There were three possible locations to set up shop: Newport Beach, Santa Ana, and Costa Mesa. We considered the population within a 10-minute drive radius of each site. All three areas are thriving, with businesses doing well and strong local activity. Each city has a high population of over 100,000 residents, with Santa Ana leading at 301,067. However, population alone does not determine the best location for business. Consumer behavior must also be considered to see who is willing to spend more on discretionary items listed in the Retail Market Potential. In the product and consumer behavior sections, Santa Ana residents outperformed the other two cities across multiple categories,

such as groceries and electronics. This pattern highlights that Santa Ana has a more active consumer market, making it an ideal location for a frozen yogurt shop.

Sustainability Analysis

Frocup came out on top with a perfect compatibility score of 1.0. All the criteria, such as the percentage of consumers who purchased frozen yogurt in the last six months, were high, combining to the maximum score. Other competitors were close behind: Afters Ice Cream scored 0.9581 and had the highest 2025 dairy-related product sales at \$17,937,200, while Bon Suisse Inc scored 0.98117, showing that the area has both strong competition and high demand. For a potential neighbor, NuYo Frozen Yogurt is a suitable choice with a middle-ground score of 0.6614, where demand exists but no single business dominates. The two lowest-scoring locations should be avoided, as demand there is likely too low to support a new shop.



Story Design & Presentation

We started by adding live Smart Map layers to show the three potential frozen yogurt locations we were considering. Next, we layered demographic data including age, income, and dessert buying habits to highlight where young adults, teenagers, and families are most likely to visit. Then we added drive time analysis maps to visualize accessibility from each location. Finally, we created a ranking table combining demand, competition, and convenience to determine the best site for a new frozen yogurt shop.

Conclusion

Based on our analysis, Frocup in Santa Ana is the most suitable location for a new frozen yogurt shop. The site benefits from a high population, a median age range that includes young adults, teenagers, and families, and a high spending index across grocery, dairy, and dessert purchases. Residents also show a strong preference for healthier desserts with fresh fruits, making this site ideal. Considering demand, demographics, accessibility, and competition, Frocup in Santa Ana offers the greatest potential for success.